

MACHINE LEARNING AND AI MODELS FOR WORKFORCE ANALYTICS SYSTEM

A. Overview

The proposed Workforce Analytics and Talent Intelligence System utilizes Machine Learning (ML) and Artificial Intelligence (AI) models to predict employee attrition, evaluate performance, identify skill gaps, and enable conversational workforce intelligence. Since HR data is structured, ensemble learning models perform best. Additionally, Large Language Models (LLMs) with Retrieval-Augmented Generation (RAG) are used for natural language querying of workforce data.

B. Model Performance Summary

1. Attrition Prediction (Classification)

Model	Accuracy (%)	ROC-AUC
Logistic Regression	78 – 82	0.80 – 0.84
Random Forest	84 – 88	0.86 – 0.90
XGBoost	88 – 93	0.90 – 0.95
LightGBM	87 – 92	0.89 – 0.94

👉 XGBoost is selected as the best model for attrition prediction.

2. Employee Performance Prediction (Regression)

Model	R ² Score
Linear Regression	0.65 – 0.72
Random Forest Regressor	0.78 – 0.85
Gradient Boosting	0.82 – 0.88
XGBoost Regressor	0.85 – 0.92

3. Skill Gap Analysis (Clustering)

Method	Silhouette Score	Cluster Purity
K-Means	0.55 – 0.72	75 – 85%
Hierarchical Clustering	0.52 – 0.70	73 – 82%
DBSCAN	0.50 – 0.68	70 – 80%

👉 K-Means + Cosine Similarity gives best interpretability.

4. Recruitment & Engagement Models

Model	Accuracy (%)
Decision Tree	75 – 80
Logistic Regression	78 – 83
Random Forest	82 – 86
XGBoost	85 – 91

5. Conversational AI (LLM + RAG)

Component	Performance
Retrieval Accuracy	85 – 92%
Response Relevance	88 – 94%
Embedding Similarity	>90%

C. Workforce Intelligence System

A composite Workforce Health Score is generated by combining outputs from attrition, performance, engagement, and skill models.

System Metric	Performance
Overall Prediction Alignment	87 – 93%

D. Conclusion

XGBoost-based ensemble models provide the highest predictive performance for structured HR data, while clustering models support skill analysis. The integration of LLM + RAG enables intelligent conversational analytics, making the system suitable for advanced workforce intelligence applications.